

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1 load the necessary packages
library(tidyverse)
library(here)
library(cowplot)
library(ggthemes)
library(ggplot2)

#verify the home directory
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#read in the datasets
```

```
ntl.chem.peterpaul.processed <- read.csv(  
  file=here(  
    "./Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),  
  stringsAsFactors = TRUE)
```

```
niwo.litter.processed <- read.csv(  
  file=here("./Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),  
  stringsAsFactors = TRUE)
```

```
#2 check the date format
```

```
ntl.chem.peterpaul.processed$sampldate <- as.Date(  
  ntl.chem.peterpaul.processed$sampldate, format = "%Y-%m-%d")
```

```
niwo.litter.processed$collectDate <- as.Date(  
  niwo.litter.processed$collectDate, format = "%Y-%m-%d")
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3 define plot theme
```

```
my_theme <- theme_base() +  
  theme(  
  
    plot.title = element_text(  
      face = "bold", size = 14),  
    axis.title = element_text(  
      size = 12, color = "black"),  
    axis.text = element_text(  
      size = 10, color = "black"),  
    axis.ticks = element_line(color = "blue"),  
    panel.grid.major = element_line(color = "grey"),  
    panel.grid.minor = element_blank(),  
    plot.background = element_rect(  
      color = "grey", fill = "grey"),  
    legend.position = 'right',  
    legend.title = element_text(color = "black")  
  
  )
```

```
theme_set(my_theme)
```

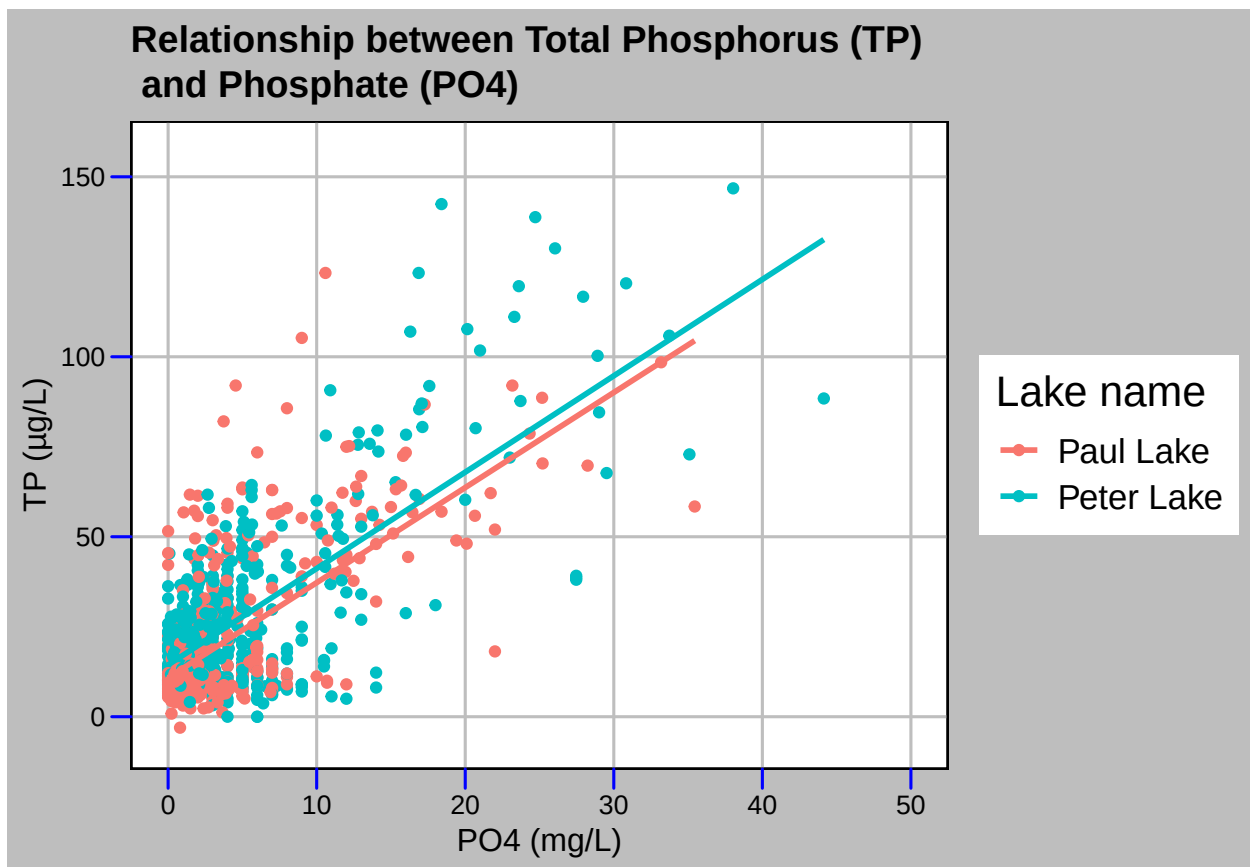
Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
p_tp_po4 <- ggplot(ntl.chem.peterpaul.processed,
                   aes(po4, tp_ug, color = lakename))+
  geom_point()+
  geom_smooth(method = "lm", se = FALSE)+
  labs(title= "Relationship between Total Phosphorus (TP) \n and Phosphate (P04)",
       x = "P04 (mg/L)", y="TP (µg/L)", color = "Lake name")+
  #adjust the limits
  xlim(0, 50)

p_tp_po4
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

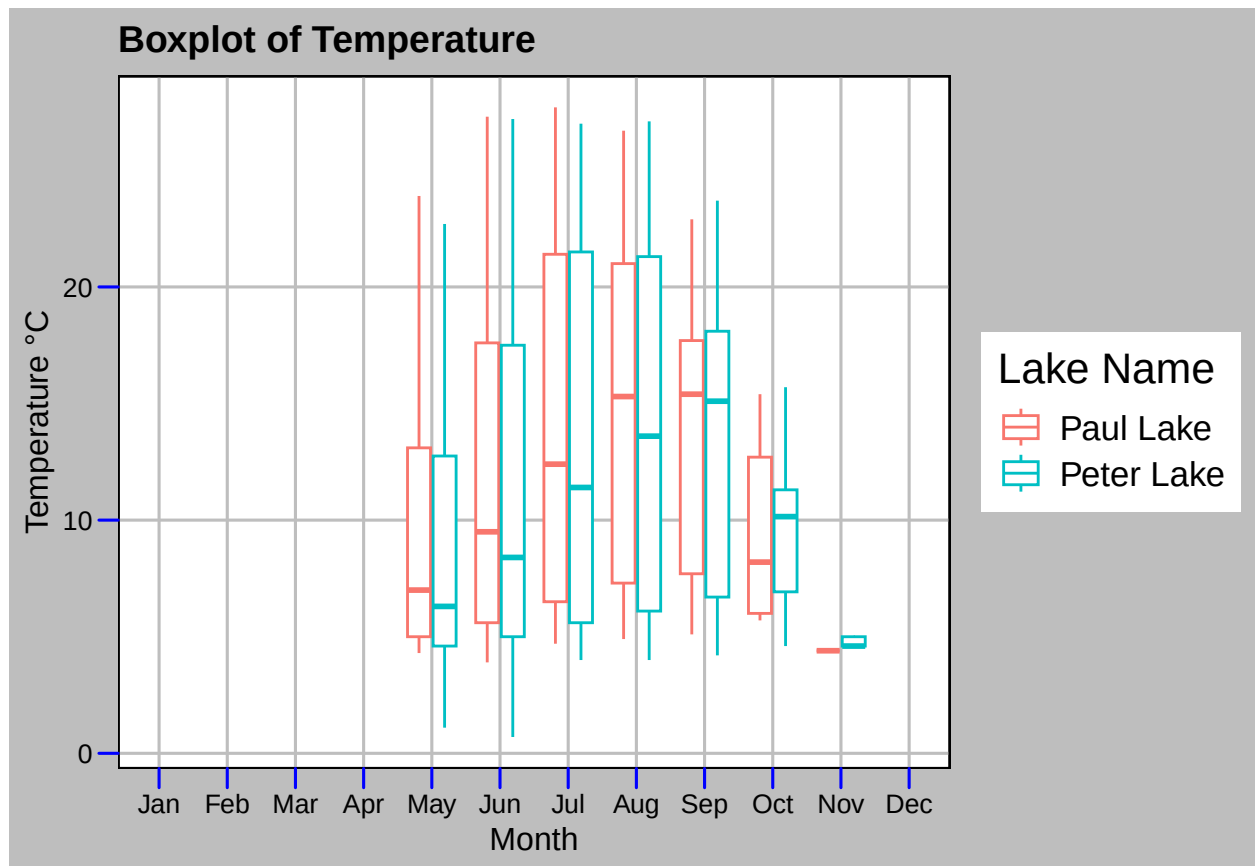
```
#5

# boxplot for temperature
p_temp <- ggplot(ntl.chem.peterpaul.processed,
  aes(x=factor(month, levels=1:12, labels = month.abb),
    y=temperature_C, color=lakename))+
  geom_boxplot()+
  labs(title = "Boxplot of Temperature", color = "Lake Name",
    x = "Month", y = "Temperature °C")+
  scale_x_discrete(name = "Month", drop = FALSE)

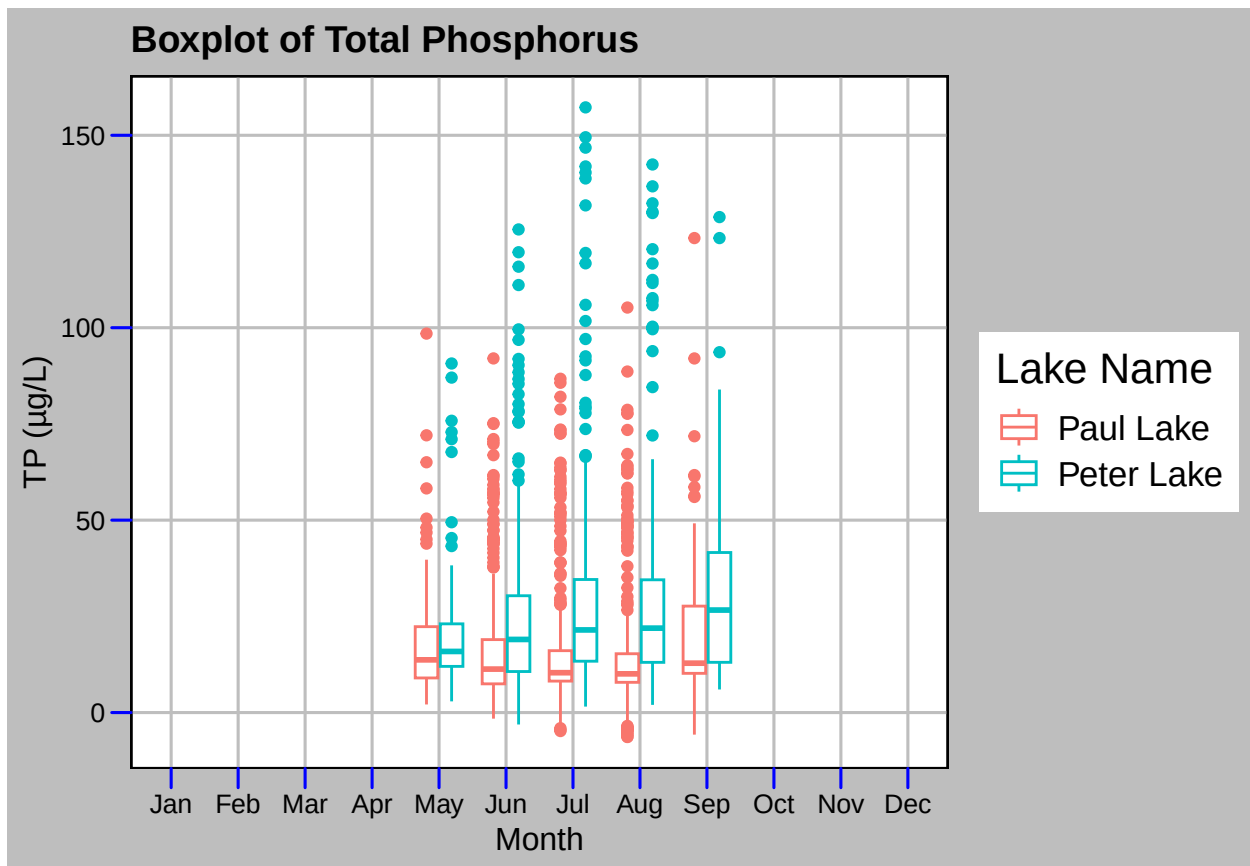
# boxplot for TP
p_tp <- ggplot(ntl.chem.peterpaul.processed,
  aes(x=factor(month, levels=1:12, labels = month.abb),
    y=tp_ug, color=lakename))+
  geom_boxplot()+
  labs(title = "Boxplot of Total Phosphorus", color = "Lake Name",
    x = "Month", y = "TP (µg/L)")+
  scale_x_discrete(name = "Month", drop = FALSE)

# boxplot for TN
p_tn <- ggplot(ntl.chem.peterpaul.processed,
  aes(x=factor(month, levels=1:12, labels = month.abb),
    y=tn_ug, color=lakename))+
  geom_boxplot()+
  labs(title = "Boxplot of Total Nitrogen", color = "Lake Name",
    x = "Month", y = "TN (µg/L)")+
  scale_x_discrete(name = "Month", drop = FALSE)

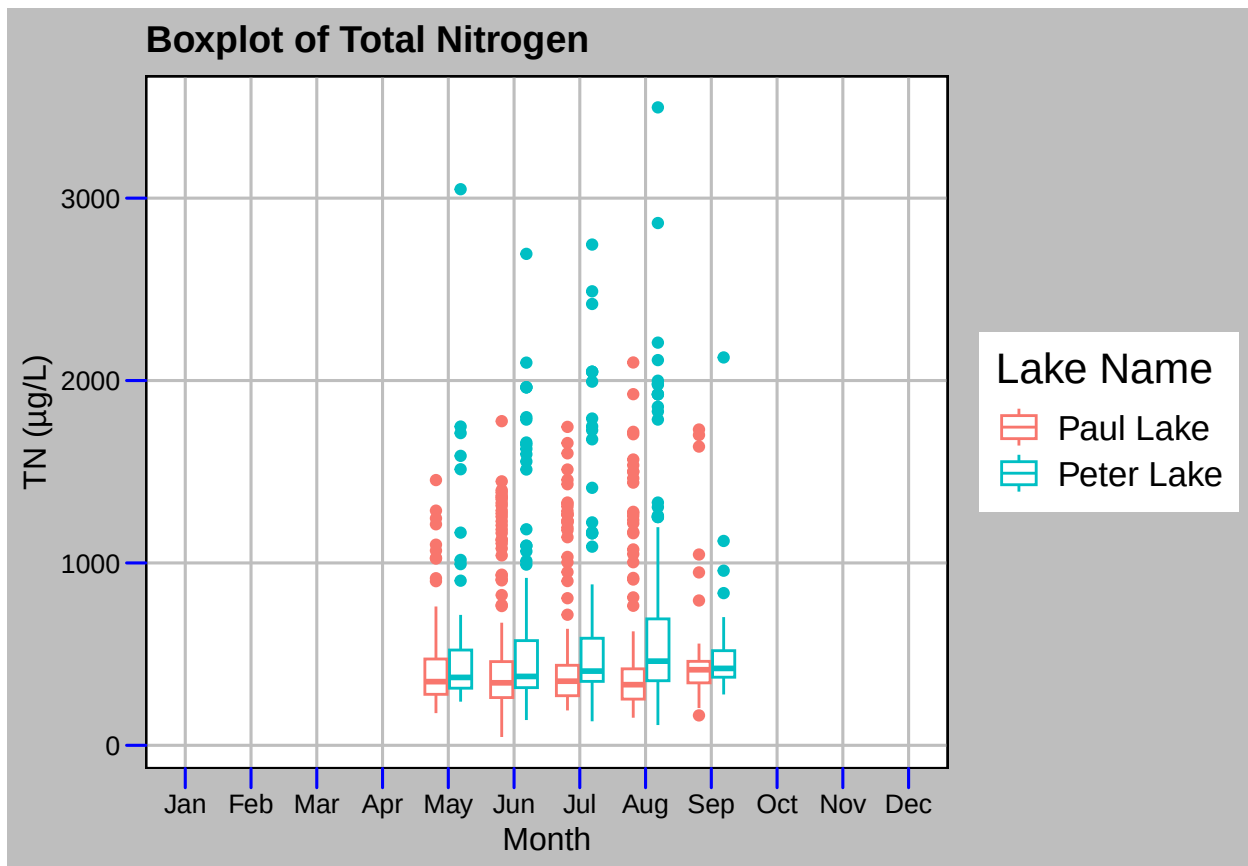
# view the individual plots
p_temp
```



p_tp

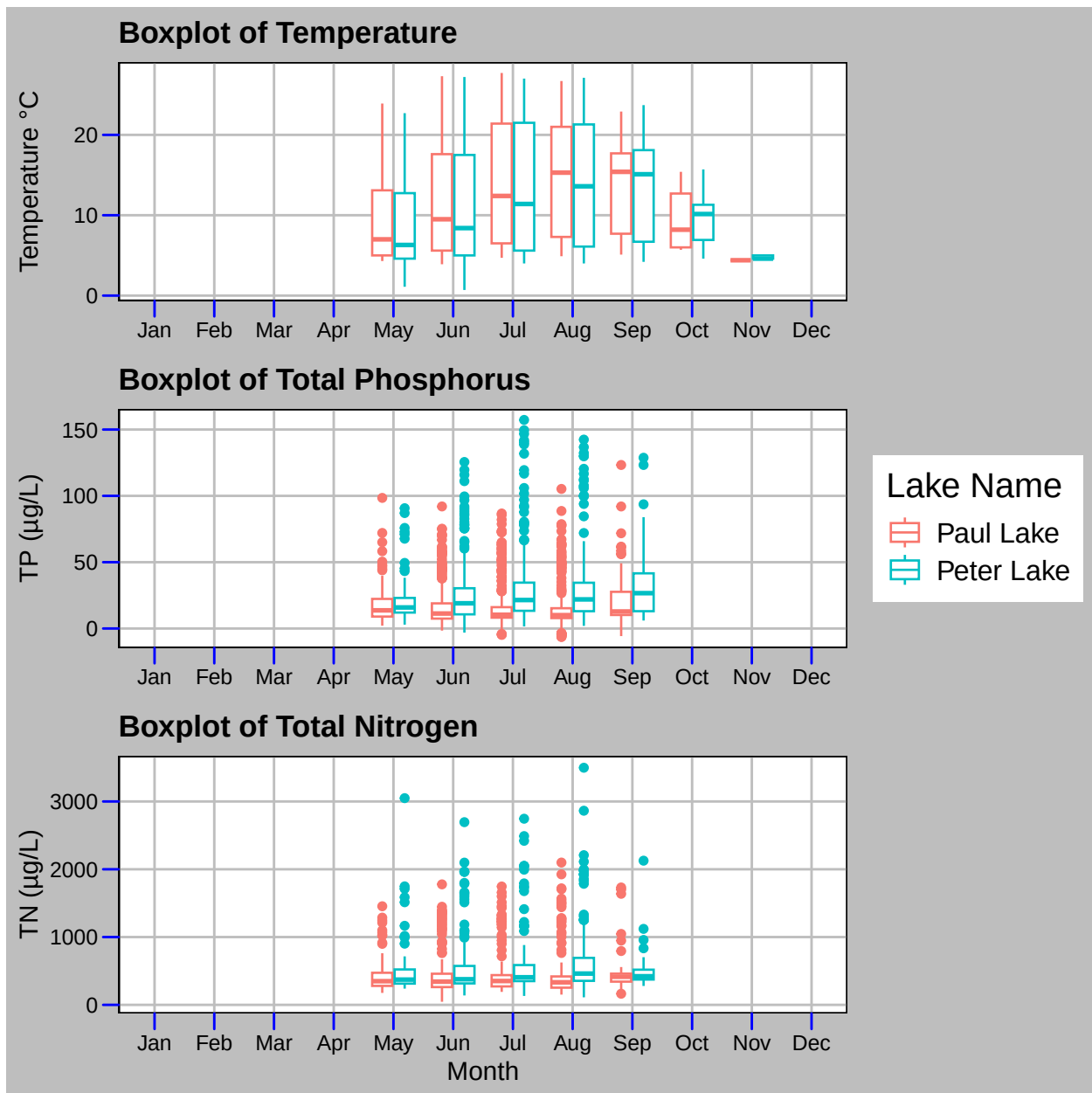


p_tn



```
# load the `cowplot` package
library(cowplot)
# edit the plots
p_temp <- p_temp+
  theme(axis.title.x = element_blank(), legend.position = "none")
p_tp <- p_tp+
  theme(axis.title.x = element_blank(), legend.position = "right")
p_tn <- p_tn+
  theme(legend.position = "none")

# combine the three plot
plot_grid(p_temp, p_tp, p_tn, nrow = 3, align = 'v', rel_heights = c(7, 7, 8))
```



Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: The temperature shows clear seasonal variation, as the lakes are warmer in summer months and cooler in winter. The median temperature of the Paul Lake is slightly higher than the Peter Lake from May to September, and overall the difference is very small. On the other hand, TP and TN both show right-skewed data distribution and fluctuate modestly with small peaks in summer (July-August). Generally, the Peter Lake has higher nutrient concentrations, and the some TN measurements might exceed the health threshold. The lack of winter data might result from the lake being frozen.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)

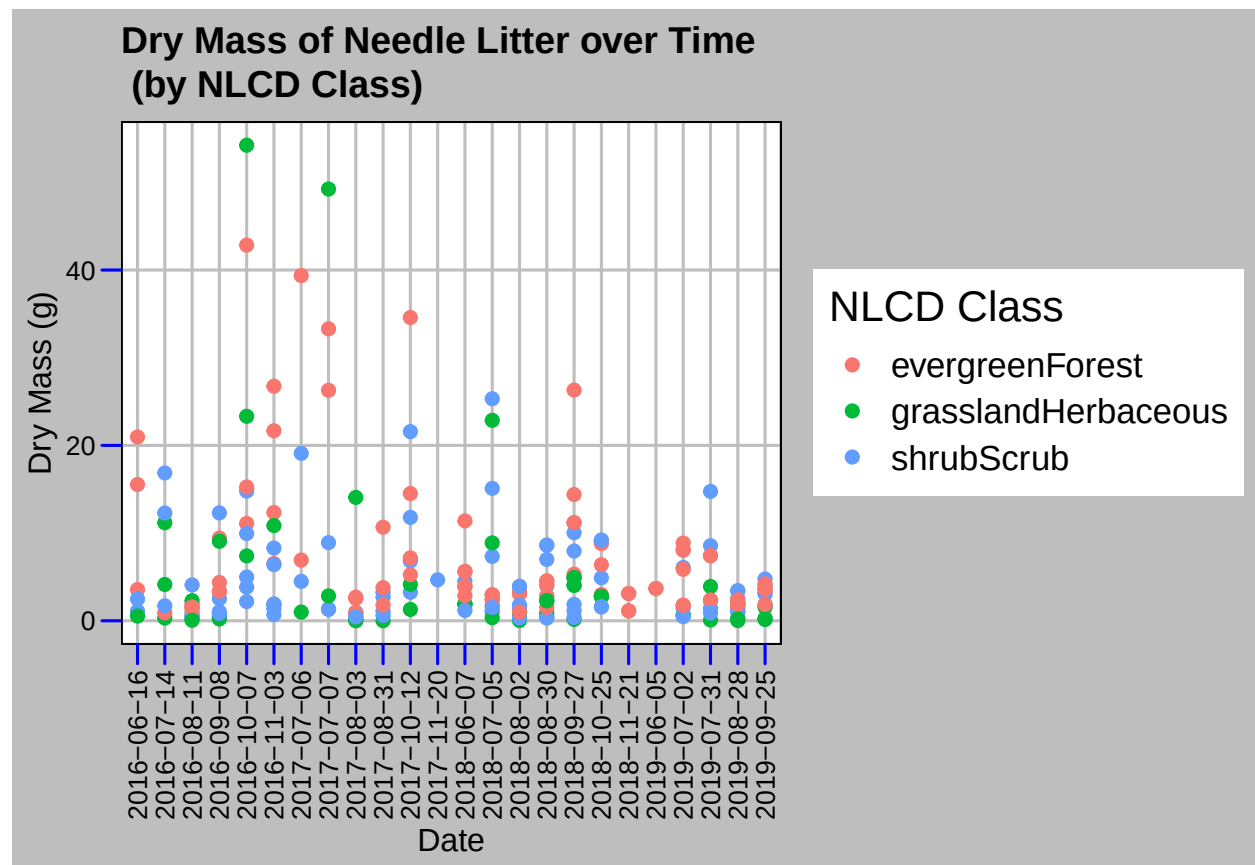
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6

# subset the dataset only displaying the "Needles" functional group
niwo.needles <- niwo.litter.processed %>%
  filter(functionalGroup == "Needles")

p6 <- ggplot(niwo.needles, aes(x = as.factor(collectDate), y = dryMass, color = nlcdClass)) +
  geom_point(size = 2) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5)) +
  labs(
    title = "Dry Mass of Needle Litter over Time \n (by NLCD Class)",
    x = "Date",
    y = "Dry Mass (g)",
    color = "NLCD Class")

p6 #view the plot
```



```
#7

p7 <- ggplot(niwo.needles, aes(x = as.factor(collectDate), y = dryMass)) +
  geom_boxplot(color = "darkgreen") +
  geom_point(size = 1, color = "darkgreen") +
```

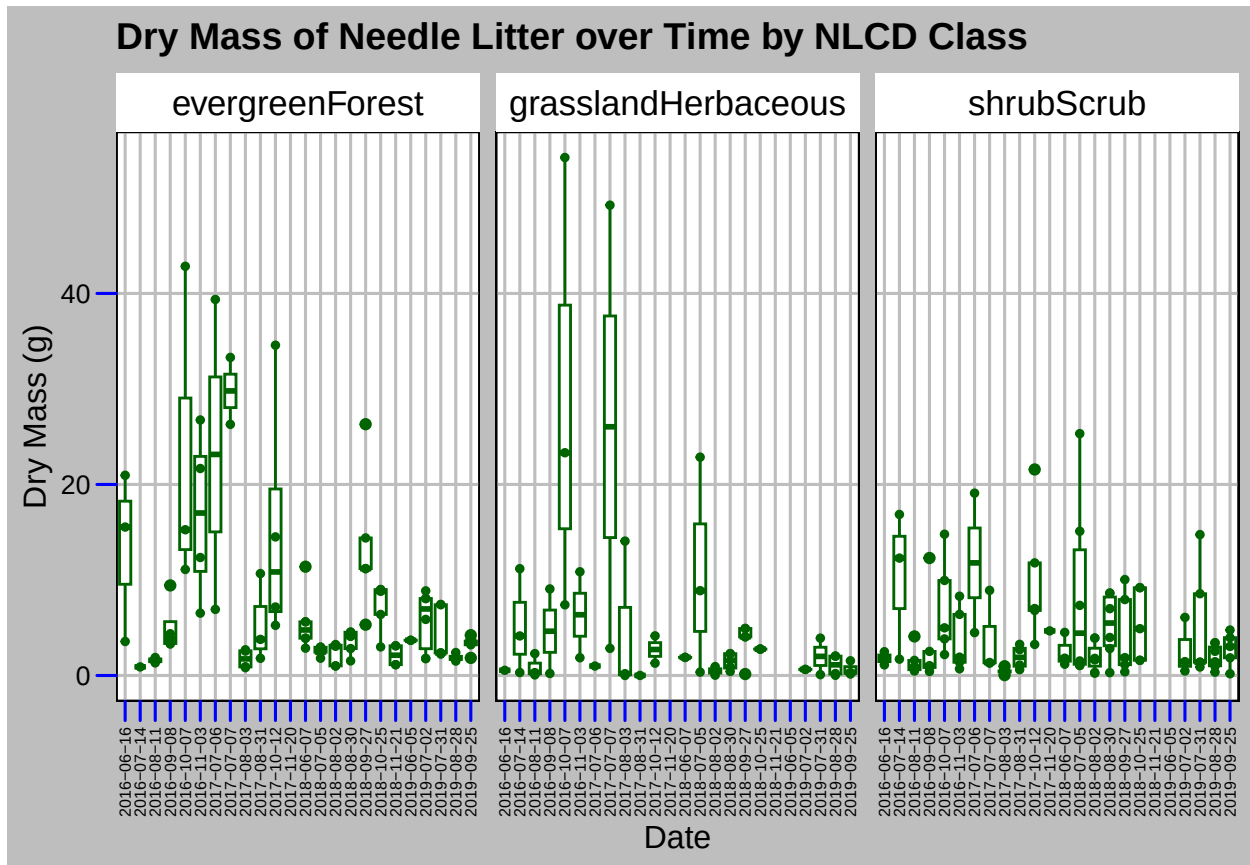
```

theme(axis.text.x = element_text(angle = 90, vjust = 0.5))+
facet_wrap(~ nlcdClass) +
labs(
  title = "Dry Mass of Needle Litter over Time by NLCD Class",
  x = "Date",
  y = "Dry Mass (g)")+
theme(axis.text.x = element_text(size=6))

```

view the plots

p7



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: The plot for Question 7 is more effective because it separates each land cover type clearly, which reduces the overlap of points from different groups and with different colors. Though the general trend is visualized in Plot 6, it is hard to see the temporal trends across NLCD classes. In Plot 7, not only the temporal trends are more obvious, but also the box plot can be applied to present additional statistics.